

Notes: Acharya, Engle, Jager & Steffen (RFS, 2024)

Why Did Bank Stocks Crash during COVID-19?

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Bank sample and stock returns	2
2.2	Credit-line exposure and liquidity risk	2
2.3	Gross and net drawdowns	3
2.4	Credit-line repayments	3
2.5	Credit-line pricing and stress-test inputs	3
3	Empirical specifications and stress-test framework	3
3.1	Baseline stock-return regression	3
3.2	Portfolio-composition controls	4
3.3	Mechanism regression: funding versus capital	4
3.4	Bank lending regressions	4
3.5	Repayment and recovery regression	4
3.6	Credit-line fee pricing	4
3.7	Stress-test extension	5
4	Identification and empirical design	5
4.1	What identifies the early-crisis effect	5
4.2	Why COVID is informative	5
4.3	Why this is not simply GFC-style funding fragility	5
4.4	What the paper cannot fully prove	5
5	Tables and figures	6
5.1	Figure 1: credit-line growth, drawdowns, and bank stock prices	6
5.2	Figure 2 and Figure 3: liquidity risk and stock-price sensitivity	6
5.3	Table 1 and Table 2: summary statistics and baseline returns	7
5.4	Table 3 and Table 4: robustness and timing	8
5.5	Figure 4 and Table 5: funding channel versus capital channel	10
5.6	Table 6 and Table 7: lending consequences and repayments	10
5.7	Table 8 and Table 9: other crises and credit-line pricing	12
5.8	Figure 5, Figure 6, and Table 10: issuance, drawdown slopes, and stress tests	13
6	Short summary	14
7	Conclusion	15
7.1	Core conclusion	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Why did bank stocks crash so severely at the start of COVID-19, and why did they partially recover after the second quarter of 2020?
- **Core answer.** A two-sided credit-line channel explains the crash and recovery: firms drew down bank credit lines in March 2020, encumbering bank capital, and later repaid them, releasing capital.
- **Why banks were exposed.** U.S. public firms' undrawn bank credit lines rose sharply before COVID, from roughly 0.7% of GDP in 2009 to 5.7% of GDP in 2019.
- **Balance-sheet liquidity risk.** The key bank-level exposure is unused commitments plus wholesale finance minus liquid assets, scaled by assets.
- **Main stock-return result.** A one-standard-deviation increase in liquidity risk reduces bank stock returns by about 8.4 percentage points from January 1 to March 23, 2020.
- **Why not just borrower risk?** The effect survives controls for bank exposure to COVID-affected industries and riskier loan portfolios.
- **Mechanism.** The funding-liquidity channel is weak during COVID because deposits flowed into banks. The capital channel is strong: credit-line drawdowns convert off-balance-sheet commitments into on-balance-sheet loans that absorb scarce capital.
- **Real effects.** Banks with larger credit-line drawdowns cut new loan originations, even though their existing exposures rose.
- **Recovery.** Credit-line repayments in 2020Q2–Q3 reduce capital encumbrance and help explain why some bank stocks recover from the March crash.
- **Stress-testing implication.** Credit lines are like banks writing put options on aggregate risk. Stress tests should include conditional credit-line drawdown risk and its contribution to systemic capital shortfall.

2 Data and measurement

2.1 Bank sample and stock returns

- The bank sample contains 147 U.S. publicly listed banks.
- Stock returns are measured over crisis windows in 2020, especially January 1–March 23, 2020, before decisive monetary and fiscal policy support had stabilized markets.
- The paper compares banks with nonfinancial firms and nonbank financial firms, and then studies the cross-section of bank returns.

Interpretation: The central cross-sectional question is whether banks that had written more credit-line liquidity insurance were punished more severely by equity markets during the COVID drawdown wave.

2.2 Credit-line exposure and liquidity risk

The paper defines balance-sheet liquidity risk as

$$\text{LiquidityRisk}_i = \frac{\text{UnusedCommitments}_i + \text{WholesaleFunding}_i - \text{Liquidity}_i}{\text{Assets}_i}. \quad (1)$$

- *Unused commitments* capture undrawn credit lines that may be converted into loans during stress.
- *Wholesale funding* captures runnable nondeposit funding.
- *Liquidity* is cash or cash-equivalent assets that can absorb liquidity shocks.

Interpretation: This is an ex ante measure of how exposed a bank is to correlated liquidity demand by borrowers and potentially fragile funding sources.

2.3 Gross and net drawdowns

To separate funding and capital channels, the paper defines

$$\text{GrossDrawdowns}_i = \frac{\Delta \text{Drawn C\&I Credit Lines}_i}{\text{Assets}_i}, \quad (2)$$

$$\text{NetDrawdowns}_i = \frac{\Delta \text{Drawn C\&I Credit Lines}_i - \Delta \text{Deposits}_i}{\text{Assets}_i}. \quad (3)$$

- Gross drawdowns capture how much credit-line exposure moves onto the balance sheet.
- Net drawdowns subtract deposit inflows, so they capture whether drawdowns create a funding shortfall.

Interpretation: If the funding channel is dominant, net drawdowns should matter. If the capital-encumbrance channel is dominant, gross drawdowns should matter even when deposits rise.

2.4 Credit-line repayments

- The paper tracks repayments in 2020Q2 and 2020Q3.
- Repayments are scaled either by total commitment or by the remaining drawn balance.
- Repayments are used to explain the recovery in bank market values after the initial crash.

Interpretation: Repayments turn drawn loans back into undrawn commitments, releasing regulatory capital and helping banks' stock values recover.

2.5 Credit-line pricing and stress-test inputs

- The paper studies credit-line fees from DealScan: all-in-spread-drawn (AISD), all-in-spread-undrawn (AISU), and up-front fees.
- It asks whether fees price banks' exposure to systemic drawdown risk.
- The stress-test exercise uses the relationship between drawdown rates and market returns to construct a stressed drawdown rate and incremental systemic risk.

Interpretation: If credit lines are put options on aggregate risk, bank regulation should treat them as contingent claims that move onto the balance sheet in bad states.

3 Empirical specifications and stress-test framework

3.1 Baseline stock-return regression

The main cross-sectional regression is

$$R_i^{\text{Jan1-Mar23,2020}} = \alpha + \beta \text{LiquidityRisk}_i + \Gamma' X_i + \varepsilon_i. \quad (4)$$

- R_i is bank i 's excess stock return over the early COVID crisis window.
- X_i includes bank controls such as beta, NPL/loans, noninterest income, size, ROA, deposits, loan/assets, idiosyncratic volatility, real estate beta, derivatives, consumer loans, credit card commitments, and SRISK/assets in some specifications.

Interpretation: A negative β means that investors viewed undrawn credit-line exposure as a risky off-balance-sheet commitment once COVID created correlated drawdown demand.

3.2 Portfolio-composition controls

A key concern is that banks with more credit lines may lend to riskier or more COVID-exposed borrowers. The paper estimates

$$R_i = \alpha + \beta \text{LiquidityRisk}_i + \delta \text{COVIDExposure}_i + \Gamma' X_i + \varepsilon_i, \quad (5)$$

using several measures of borrower exposure to COVID-affected industries.

Interpretation: The liquidity-risk coefficient remains large and negative, so the crash is not simply a borrower-composition story.

3.3 Mechanism regression: funding versus capital

The mechanism tests relate stock returns to gross and net drawdowns:

$$R_i = \alpha + \beta_1 \text{GrossDrawdowns}_i + \beta_2 \text{NetDrawdowns}_i + \text{eta}_3 \text{CapitalBuffer}_i + \text{extinteractions} + \Gamma' X_i + u_i. \quad (6)$$

Interpretation: The paper finds that gross drawdowns matter more than net drawdowns. This points to capital encumbrance, not a shortage of deposit funding, as the main channel.

3.4 Bank lending regressions

The lending tests use a difference-in-differences style panel specification:

$$Y_{i,t} = \alpha_i + a u_t + \text{eta}(\text{DrawdownExposure}_i \times \text{Post}_t) + \Gamma' X_{i,t} + u_{i,t}, \quad (7)$$

where $Y_{i,t}$ is either loan exposures or new loan originations.

Interpretation: Drawdowns increase exposures but reduce new originations. This is the real-economy side of the capital channel.

3.5 Repayment and recovery regression

To study the stock-price recovery, the paper estimates regressions of the form

$$\text{Recovery}_i = \alpha + \text{eta} \text{Repayments}_i + \delta \text{Fees}_i + \theta \text{MVLoss}_i + \text{interactions} + \varepsilon_i. \quad (8)$$

Interpretation: Repayments are positively related to recovery, especially for banks that lost more market value during the crash.

3.6 Credit-line fee pricing

The fee regressions are

$$\text{Fee}_{b,f,t} = \alpha + \beta \text{BankRisk}_b + \Gamma' \text{BankControls}_b + \Pi' \text{FirmControls}_f + FE + \varepsilon_{b,f,t}. \quad (9)$$

Interpretation: Fees price some bank risk, but the evidence does not suggest that credit-line fees fully compensate banks for aggregate drawdown risk.

3.7 Stress-test extension

The stress-test calculation starts from

$$DrawdownRate_t = \alpha + \lambda MarketReturn_t + u_t, \quad (10)$$

then predicts a drawdown rate under a severe market correction. The incremental capital shortfall from credit-line drawdowns is

$$Incremental\ SRISK_i^{CL} = DrawdownRate \times 8\% \times UnusedC\&ICommitments_i. \quad (11)$$

Interpretation: Stress tests should add a credit-line drawdown module because off-balance-sheet commitments become capital-absorbing loans in aggregate stress states.

4 Identification and empirical design

4.1 What identifies the early-crisis effect

- The main variation is bank-level exposure to unused commitments and liquidity risk before the COVID shock fully materialized.
- The early window ends March 23, 2020, before the full effect of policy interventions and repayments.
- The paper controls for many bank fundamentals and borrower-industry exposures.

Interpretation: The design asks whether banks with larger ex ante contingent liquidity commitments were more severely marked down when the aggregate drawdown state arrived.

4.2 Why COVID is informative

- COVID created a rare, highly correlated demand for credit-line drawdowns.
- Bond markets froze, especially for riskier firms, so borrowers drew on prearranged bank credit.
- Deposit inflows were strong, allowing the paper to separate funding stress from capital encumbrance.

Interpretation: COVID is a strong test of the credit-line channel because drawdown demand was both large and aggregate.

4.3 Why this is not simply GFC-style funding fragility

- During the GFC, wholesale funding fragility was central.
- During COVID, banks received deposit inflows, so funding was less binding.
- The cross-section points to gross drawdowns and capital buffers rather than wholesale funding.

Interpretation: The same instrument—a corporate credit line—can stress banks through different channels in different crises.

4.4 What the paper cannot fully prove

- Bank credit-line exposure is not randomly assigned.
- Some borrower risk may remain unobserved despite extensive controls.
- The analysis focuses on public banks and observed credit-line data, not every bank-borrower contract.
- The stress-test exercise is a policy proposal calibrated from historical and COVID drawdown behavior, not a fully structural model of bank regulation.

Interpretation: The paper’s strength is the convergence of evidence: stock returns, drawdowns, lending, repayments, pricing, and stress-test calculations all point to the credit-line channel.

5 Tables and figures

5.1 Figure 1: credit-line growth, drawdowns, and bank stock prices

Read:

- Panel A shows the pre-COVID expansion of bank credit-line financing to U.S. public firms.
- Panels B and C show that drawdowns surged in March 2020 and were concentrated among BBB, non-investment-grade, and unrated borrowers.
- Panel D shows banks crashing more severely and recovering more slowly than nonfinancial and nonbank financial firms.

Economic message: The credit-line shock is both large and aggregate. The same commitments that provide insurance to firms create a correlated exposure for banks.

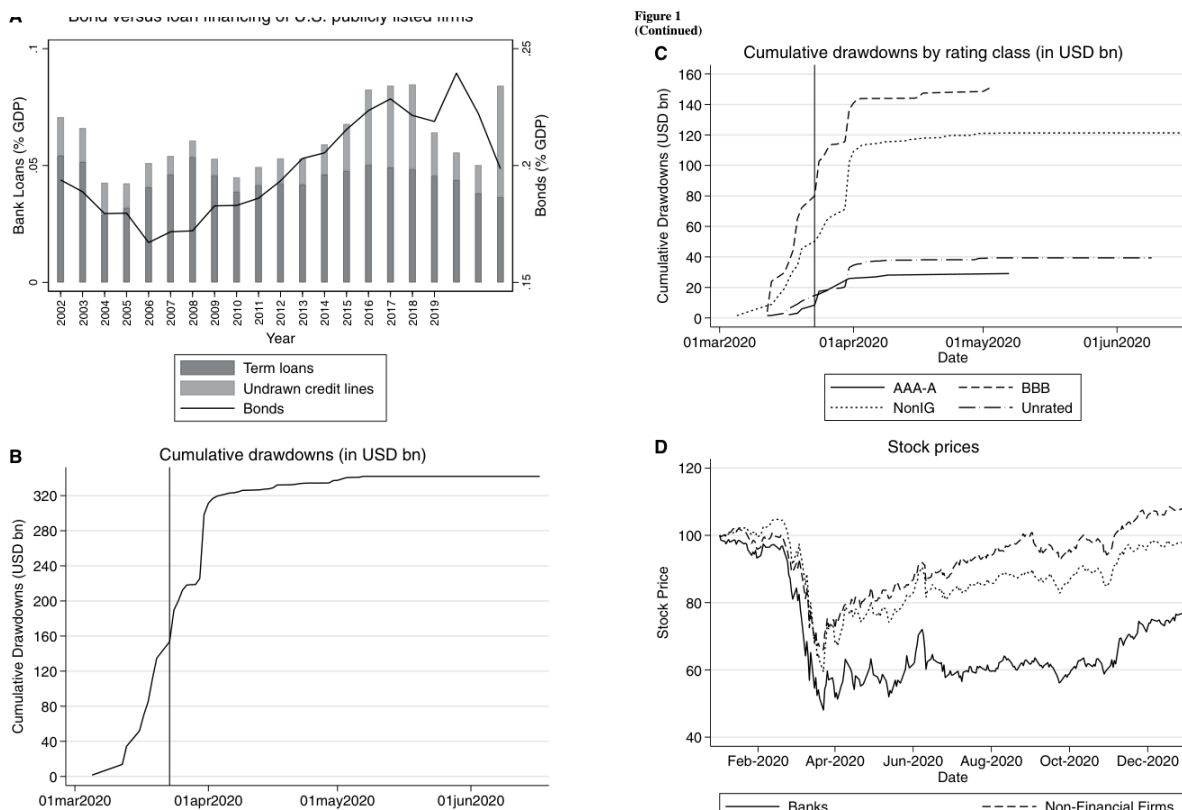


Figure 1, panels A–B

Figure 1, panels C–D

5.2 Figure 2 and Figure 3: liquidity risk and stock-price sensitivity

Read:

- Figure 2 shows that liquidity risk declines over the post-GFC period but rises again before COVID as unused C&I commitments grow while wholesale funding falls.
- Figure 3 shows that high-liquidity-risk banks underperform low-liquidity-risk banks during 2020, and the cross-sectional relation between liquidity risk and March stock returns is negative.

Economic message: Markets distinguish banks by their exposure to contingent credit-line drawdowns.

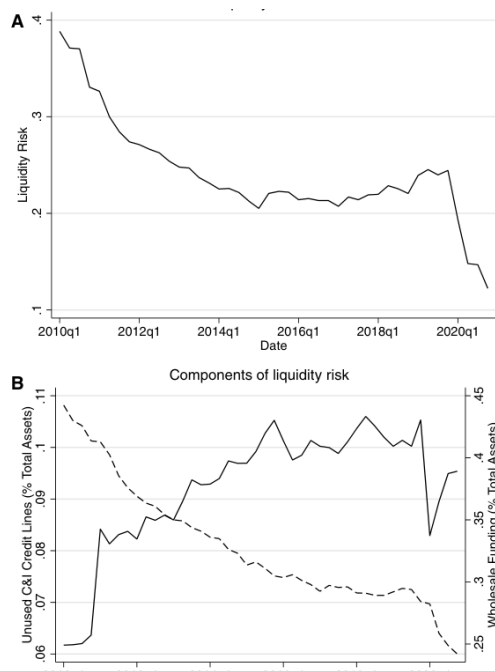


Figure 2: balance-sheet liquidity risk

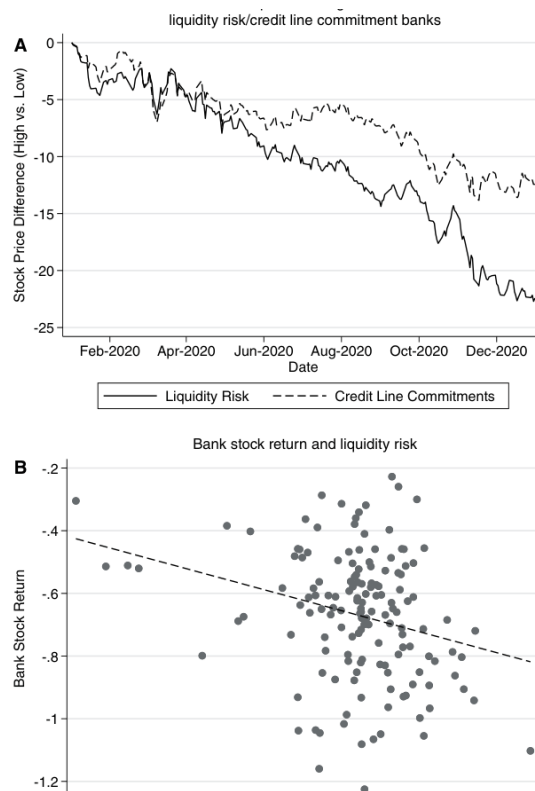


Figure 3: bank stocks and liquidity risk

5.3 Table 1 and Table 2: summary statistics and baseline returns

Read:

- Table 1 shows that bank stock returns were sharply negative: mean return is -7.2% in January, -12.5% in February, -47.2% from March 1 to March 23, and -66.9% from January 1 to March 23.
- Mean liquidity risk is 0.195, mean unused commitments/assets is 0.077, and mean liquidity/assets is 0.136.
- Table 2 shows a strong negative coefficient on liquidity risk in every specification. In columns 5–6, the coefficient is about -0.57 and -0.55, both significant.
- Adding SRISK/assets raises R^2 to about 0.50 but leaves the liquidity-risk coefficient large and significant.

Economic message: Bank liquidity risk is not a weak proxy. It explains a substantial share of the cross-sectional crash in bank equity values.

Descriptive statistics					
Variable	Obs.	Mean	Std. dev.	Min	Max
<i>A. Bank stock returns</i>					
Return, January 2020	147	-0.072	0.046	-0.181	0.064
Return, February 2020	147	-0.125	0.040	-0.246	0.071
Return, January 1 to March 23, 2020	147	-0.472	0.186	-1.084	-0.131
Return, January 1 to March 23, 2020	147	-0.669	0.206	-1.225	-0.227
<i>B. Bank characteristics</i>					
<i>Liquidity risk</i>	147	0.195	0.147	-0.453	0.590
<i>Unused LC / assets</i>	147	0.077	0.051	0.000	0.263
<i>Liquidity / assets</i>	147	0.136	0.109	0.029	0.607
<i>Wholesale funding / assets</i>	147	0.144	0.100	0.013	0.624
<i>Beta</i>	147	1.170	0.328	0.156	2.313
<i>NPL / loans</i>	147	0.008	0.008	0.000	0.044
<i>Noninterest income</i>	147	0.268	0.185	0.021	0.966
<i>log(Assets)</i>	147	16.982	1.437	14.397	21.712
<i>ROA</i>	147	0.013	0.006	0.003	0.061
<i>Deposits / loans</i>	147	1.306	1.130	0.504	11.002
<i>Income diversity</i>	147	0.446	0.212	0.043	0.993
<i>Z-score</i>	147	3.619	0.536	1.859	5.060
<i>Loans / assets</i>	147	0.670	0.166	0.027	0.899
<i>Deposits / assets</i>	147	0.745	0.105	0.191	0.879
<i>Idiosyncratic volatility</i>	147	0.200	0.041	0.121	0.417
<i>Real estate beta</i>	147	0.544	0.197	-0.266	1.136
<i>Primary dealer</i>	147	0.041	0.199	0.000	1.000
<i>Derivatives / assets</i>	147	1.161	4.753	0.000	37.242
<i>Credit card commitments / assets</i>	147	0.075	0.389	0.000	3.998
<i>Consumer loans / assets</i>	147	0.056	0.117	0.000	0.828
<i>SRISK / assets</i>	147	0.003	0.007	0.000	0.039

Table 1: descriptive statistics

Liquidity risk and bank stock returns						
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Liquidity risk</i>	-0.329*** (.000)	-0.409*** (.000)	-0.565*** (.000)	-0.550*** (.000)	-0.568*** (.000)	-0.551*** (.000)
<i>Equity beta</i>	0.734*** (.000)	0.706*** (.000)	0.566*** (.001)	0.557*** (.001)	0.577*** (.001)	0.476*** (.004)
<i>NPL / loans</i>		-7.038*** (.000)	-3.682** (.033)	-3.603** (.039)	-3.408* (.054)	-3.665** (.035)
<i>Equity ratio</i>		0.522 (.425)	-0.119 (.858)	-0.103 (.878)	-0.519 (.443)	-0.897 (.179)
<i>Noninterest income</i>		0.297*** (.003)	0.169 (.139)	0.189 (.106)	0.132 (.273)	0.0973 (.412)
<i>log(Assets)</i>		-0.000996 (.938)	-0.0330** (.046)	-0.0363** (.036)	-0.0210 (.267)	0.00422 (.844)
<i>ROA</i>		-3.726 (.310)	1.193 (.757)	1.167 (.766)	5.406 (.237)	6.158 (.163)
<i>Deposits / loans</i>		-0.0217 (.115)	-0.057*** (.001)	-0.054*** (.002)	-0.015*** (.002)	-0.054*** (.003)
<i>Income diversity</i>			-0.0226 (.799)	-0.0343 (.705)	-0.0257 (.775)	-0.0263 (.747)
<i>Distance-to-default</i>			0.0606* (.061)	0.0581* (.075)	0.0583* (.067)	0.0517* (.075)
<i>Loans / assets</i>			-0.483** (.020)	-0.461** (.032)	-0.408* (.062)	-0.352* (.099)
<i>Deposits / assets</i>			-0.0587 (.786)	-0.0207 (.938)	-0.0873 (.735)	-0.235 (.346)
<i>Idiosyncratic volatility</i>			-1.174*** (.003)	-1.206*** (.002)	-1.018** (.017)	-1.051** (.014)
<i>Real estate beta</i>			0.180* (.099)	0.184* (.093)	0.113 (.380)	0.0951 (.441)
<i>Current primary dealer indicator</i>				0.0845 (.430)	0.00641 (.958)	-0.0951 (.381)
<i>Derivatives / assets</i>				-0.00151 (.808)	-0.000340 (.958)	0.00526 (.415)
<i>Credit card commitments / assets</i>					-0.0371 (.510)	-0.0926 (.135)
<i>Consumer loans / assets</i>					-0.218 (.395)	-0.147 (.591)
<i>SRISK / assets</i>						-6.409*** (.009)
<i>R-squared</i>	.256	.354	.448	.449	.462	.502
<i>Number obs.</i>	147	147	147	147	147	147

This table reports the results of OLS regressions of U.S. banks' excess stock returns over the January 1, 2020, to March 23, 2020, period on bank *Liquidity risk* and a bank's *Equity beta* and control variables. *Equity beta* is constructed as bank stock beta relative to the S&P 500 using daily stock returns over the 2019 period, multiplied

Table 2: liquidity risk and bank stock returns

5.4 Table 3 and Table 4: robustness and timing

Read:

- Table 3 controls for multiple measures of exposure to COVID-affected industries. Liquidity risk remains negative and statistically significant across specifications.
- Table 4 shows that the strongest effect appears in the early COVID window and especially around March 2020.
- Unused C&I commitments are the main component of liquidity risk; wholesale funding is not central during COVID.
- After policy interventions, liquidity risk generally loses explanatory power, except for a weaker reappearance in 2020Q3.

Economic message: The result is episodic and tied to realized drawdown risk, not just general bank fragility.

Table 3
Controlling for bank portfolio composition via exposure to COVID-19-affected industries

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Liquidity risk</i>	−0.568*** (.000)	−0.543*** (.000)	−0.546*** (.000)	−0.527*** (.000)	−0.481*** (.000)	−0.530*** (.000)
<i>Affected industries</i> (β_{COVID})	−1.410*** (.005)	−0.531* (.097)	−0.455 (.116)	−0.526*** (.005)	−0.635*** (.000)	−0.493** (.026)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Affected measure	Fahlenbrach, Rageth, and Stulz 2021, stock performance	Moody's 2020, COVID industries	Koren and Peto 2020, customer share	Dingel and Neiman 2020, telework	Fahlenbrach, Rageth, and Stulz 2021, 6 NAIC level COVID industries	Koren and Peto 2020, presence share
R-squared	.505	.475	.475	.502	.537	.498
Number obs.	147	147	147	147	147	147
	(7)	(8)	(9)	(10)	(11)	(12)
<i>Liquidity risk</i>	−0.515*** (.000)	−0.518*** (.000)	−0.541*** (.000)	−0.524*** (.000)	−0.534*** (.000)	−0.521*** (.000)
<i>Affected industries</i> (β_{COVID})	−0.541** (.013)	−0.709*** (.004)	−0.221* (.090)	−0.910** (.018)	−1.528*** (.001)	−2.090*** (.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Affected measure	Koren and Peto 2020, teamwork share	YoY sales decline	Chodorow-Reich et al. 2022, abnormal employment decline	O*NET, physical proximity	O*NET, face-to-face discussion	O*NET, external customers
R-squared	.496	.519	.476	.501	.517	.504
Number obs.	147	147	147	147	147	147

Panel A reports the results of OLS regressions of U.S. banks' excess stock returns over March 1, 2020, to March 23, 2020, on banks' *Liquidity risk*. Columns 1–12 add different measures that proxy for bank exposures to COVID-19-affected industries. *Affected industries* (β_{COVID}) are calculated in regressions of bank excess stock returns on stock returns of COVID-19-affected industries and various (macro) variables: *Market return*, *SMB*, *HML*, *risk-free interest rate*, *VIX*, *term spread*, *BBB-AAA spread*, the *Consumer price inflation* (as explained in Note at the bottom of this table). Column 13 uses the first principal component based on all 12 exposure betas. Column 14 uses a bank's average DealScan syndicated loan exposure to affected industries based on different definitions relative to total assets (*Loan exposure / assets*). All variables are defined in Appendix C, and all measures are defined in Appendix D. * $p < .1$; ** $p < .05$; *** $p < .01$ (based on robust standard errors).

Table 3: portfolio-composition controls

Liquidity risk and bank stock returns: Robustness tests**A. Liquidity risk and bank stock returns by month**

	(1) January 2020	(2) January 2020	(3) February 2020	(4) February 2020	(5) January 1, 2020, to March 23, 2020	(6) January 1, 2020, to March 23, 2020
<i>Liquidity risk</i>	−0.0594** (.022)	−0.0625** (.023)	−0.0470 (.306)	−0.0439 (.357)	−0.462*** (.000)	−0.445*** (.000)
<i>Equity beta</i>	0.0452 (.253)	0.0699* (.066)	0.0350 (.185)	0.0197 (.465)	0.497*** (.003)	0.386** (.011)
<i>SRISK / assets</i>		1.317** (.048)	−1.122* (.075)		−6.604*** (.007)	
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.341	.387	.258	.285	.413	.471
Number obs.	147	147	147	147	147	147

B. Components of liquidity risk

	(1)	(2)	(3)	(4)	(5)
	January 1, 2020, to March 23, 2020				
<i>Unused C&I loans / assets</i>	−1.110*** (.001)			−1.006*** (.001)	−1.084*** (.001)
<i>Liquidity / assets</i>		0.563*** (.004)		0.477*** (.009)	0.488*** (.006)
<i>Wholesale funding / assets</i>			−0.114 (0.562)		−0.279 (0.107)
<i>Equity beta</i>	−0.578*** (.004)	−0.513** (.012)	−0.498** (.015)	0.599*** (.004)	0.597*** (.003)
<i>SRISK / assets</i>	−6.559** (.015)	−6.733*** (.005)	−7.128*** (.005)	−6.208** (.014)	−5.922** (.018)
Controls	Yes	Yes	Yes	Yes	Yes
R-squared	.456	.439	.408	.479	.486
Number obs.	147	147	147	147	147

(Continued)

(Continued)**C. Wholesale funding and bank stock returns during COVID**

	(1)	(2)	(3)	(4)
<i>Liquidity risk</i>	−0.445*** (.000)			
<i>Unused commitments / assets</i>		−1.084*** (.001)	−1.020*** (.001)	−1.149*** (.000)
<i>Liquidity / assets</i>		0.488*** (.006)	0.487*** (.008)	0.326* (.083)
<i>Wholesale funding / assets</i> (Acharya and Mora 2015)		−0.279 (.107)		
<i>Wholesale funding / assets</i> (Dubios and Lambertini 2018)			−0.0788 (.689)	
<i>Large time deposits / assets</i>				−1.164** (.034)
<i>Foreign deposits / assets</i>				−0.0464 (.846)
<i>Subordinated debt / assets</i>				−1.581 (.445)
<i>Fed funds purchased / assets</i>				1.681 (.117)
<i>Other borrowed money / assets</i>				0.0778 (.892)
R-squared	.471	.486	.480	.523
Number obs.	147	147	147	147

D. Descriptive statistics of bank stock returns over the quarters of 2020

Variable	Obs.	Mean	Std. dev.	Min	Max
2020Q1	147	−0.511	0.181	−0.996	−0.075
2020Q2	146	0.096	0.149	−0.398	0.537
2020Q3	145	−0.079	0.104	−0.282	0.249
2020Q4	144	0.346	0.115	0.014	0.706
Total	582	−0.039	0.343	−0.996	0.706

(Continued)

Table 4, panels A–B

Table 4, panels C–D

(Continued)

E. Liquidity risk and bank stock returns after the policy interventions of March 2020

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Q2-Q4 2020			Q2 2020	Q3 2020	Q4 2020
Liquidity risk	0.0104 (.856)	-0.0406 (.446)			-0.00979 (.931)	-0.132* (.073)	-0.0368 (.714)
Unused C&I loans / assets			-0.105 (.481)	-0.194* (.094)			
Liquidity / assets			-0.0726 (.352)	0.00860 (.901)			
Wholesale funding / assets			-0.0845 (.268)	-0.101 (.148)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE		Yes		Yes			
Cluster	Bank	Bank	Bank	Bank			
R-squared	.122	.751	.123	.751	.434	.380	.441
Number obs.	435	435	435	435	146	145	144

Panel A reports the results of OLS regressions of U.S. bank's realized stock returns during January 2020 (columns 1 and 2), February 2020 (columns 3 to 4), and March 1–23, 2020 (columns 5 to 6). Regressions with control variables are based on column 5 in Table 2. Panel B reports the results of OLS regressions of U.S. banks' excess stock returns over the January 1, 2020, to March 23, 2020, period on the different components of *Liquidity risk* with control variables as in column 5 in Table 2. We first show each component separately in columns 1–3 and then add them sequentially in columns 4 and 5. Panel C reports the results of OLS regressions of U.S. banks' excess stock returns over the January 1, 2020, to March 23, 2020, period on the different components of *Liquidity Risk* and different proxies for wholesale funding with control variables as in column 5 in Table 2. Columns 1 to 5 sequentially add additional components and proxies. Panel D reports descriptive statistics of bank excess stock returns for Q1–Q4 2020. Panel E reports the results of OLS regressions of U.S. banks' excess stock returns over the Q2 to Q4 2020 period on bank *Liquidity risk*, *equity beta*, and control variables as shown in column 5 of Table 2. Control variables are lagged by one quarter. Columns 1 and 2 report the results using Liquidity Risk

Table 4, panel E: post-intervention returns

5.5 Figure 4 and Table 5: funding channel versus capital channel

Read:

- Figure 4 shows a spike in gross drawdowns during 2020Q1, while net drawdowns are negative because deposit inflows offset drawdowns.
- Table 5 shows that gross drawdowns strongly reduce bank stock returns; net drawdowns are insignificant.
- Interactions with high capital or capital buffers are positive, meaning better-capitalized banks are less hurt by drawdowns.

Economic message: The main mechanism is not a funding shortage. It is capital encumbrance: drawn credit lines absorb scarce equity capital.

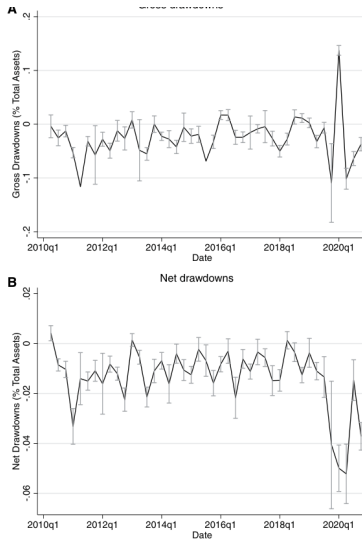


Table 5
Understanding the mechanisms: Funding versus capital during Q1 2020 (prior to policy interventions)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Net drawdowns	0.0686 (.881)		0.356 (.421)	0.393 (.333)	0.382 (.363)	0.357 (.398)	0.366 (.538)	0.305 (.469)	0.366 (.527)	0.295 (.461)
Gross drawdowns		-5.142*** (.009)	-5.618*** (.003)	-5.357*** (.007)	-9.156*** (.001)	-5.213*** (.005)	-5.615*** (.002)	-5.551*** (.003)	-9.153*** (.001)	-5.117*** (.006)
SRISK / assets				-6.236** (.039)						
Gross drawdowns × High capital					5.927** (.034)				5.913** (.033)	
Gross drawdowns × Capital buffer						1.840** (.046)				1.909** (.035)
Net drawdowns × High capital							0.186 (.845)		0.0356 (.969)	
Net drawdowns × Capital buffer								-0.115 (.454)		-0.139 (.324)
High capital					0.0298 (.559)		0.0671 (.132)		0.0304 (.554)	
Capital buffer						-1.375* (.094)		-0.697 (.377)		-1.676* (.065)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.377	.411	.415	.457	.439	.435	.425	.418	.439	.439
Number obs.	147	147	147	147	147	147	147	147	147	147

Panel A reports the results of OLS regressions of U.S. bank's excess stock returns during the January 1, 2020, to March 23, 2020, period on *Net drawdowns* (column 1) and *Gross drawdowns* (column 2) and control variables. *Net drawdowns* are defined as the change in a bank's off-balance-sheet unused C&I loan commitments minus the change in deposits (all measured during Q1 2020) relative to total assets. *Gross drawdowns* is the percentage change in a bank's off-balance-sheet unused C&I loan commitments (measured during Q1 2020). Column 4 adds *SRISK/assets* as additional control. *SRISK* is only available for banks in the NYU Stern School of Business VLAB database at vlab.stern.nyu.edu/risk. These regressions include a dummy for banks for whom we do not find *SRISK* (unreported coefficient). Column 5 includes an interaction term of *Gross drawdowns* with *High Capital*, and indicator variable that is one if a bank's equity capital ratio is above the median of the distribution. Column 6 includes an interaction term of *Gross drawdowns* with *Capital buffer*, which is the difference between a bank's equity capital ratio and the average capital ratio of all sample banks. The secular term *Capital buffer* is thus absorbed. Column 7 (column 8 include interaction terms of *Net drawdowns* and *High capital* (*Capital buffer*)). In columns 9 and 10, we compare both interaction terms of *Gross* and *Net drawdowns*. Panel B reports the results using *Deposit inflows*, defined as deposit inflows in Q1 2020 relative to total assets, instead of *Net drawdowns*. Control variables as in column 5 in Table 2 are included. All variables are defined in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (based on robust standard errors).

Figure 4: gross and net drawdowns

Table 5: funding versus capital in 2020Q1

5.6 Table 6 and Table 7: lending consequences and repayments

Read:

- Table 6 shows that drawdown-exposed banks reduce new loan originations during COVID, even though their exposures rise.
- Table 7 shows that firms repay large fractions of drawn balances in 2020Q2 and 2020Q3.
- Repayments are positively associated with bank stock recovery, especially for banks that had suffered larger market-value losses.

Economic message: Credit-line drawdowns have real intermediation consequences, and repayments reverse part of the damage by freeing bank capital.

Table 6
Implications for bank lending during the COVID-19 pandemic

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Exposures				New originations			
Gross drawdowns (gross) × Post	0.160 (.611)				-1.364 (.325)			
Net drawdowns (net) × Post	0.0299 (.579)				-0.317 (.228)			
High gross × Post	0.00115 (.849)	0.0140* (.070)	0.0131* (.075)		-0.0426** (.011)	-0.067*** (.007)	-0.0417* (.086)	
High net × Post	0.00554 (.366)	-0.00060 (.940)	-0.00303 (.648)		-0.0130 (.135)	-0.0334 (.345)	-0.0108 (.216)	
High gross × Post × Term loan indicator		-0.0363*** (.009)	-0.0366*** (.009)			0.0158 (.00626)	0.0240 (.0134)	
High net × Post × Term loan indicator		0.0173 (.229)	0.0174 (.229)				0.0134 (.470)	
Controls	No	No	No	Yes	No	No	No	Yes
Borrower × Time × Loan Type FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Borrower FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.0%	.0%	.0%	.0%	.0%	.0%	.0%	.0%
Number obs.	340,641	340,641	340,641	296,779	6,743	6,743	6,743	6,743

This table provides results of difference-in-differences regression of the change in the outstanding loan amounts (exposures) and new loan originations in the pre- versus post-COVID-19 period on *Gross* and *Net* drawdowns. The analysis is based on exposures / originations in the period between January 2019 and October 2020 (*Pre*) as the period starting April 1, 2020. Columns 1 to 4 show the results using quarterly exposures (defined as all previously issued and nonmatured credits—both term loan and credit line—reported in DealScan as the dependent variable). *High gross* (*Net*) are indicator variables equal to one if drawdowns are in the upper quartile of the distribution. *Term loan indicator* is an indicator variable equal to one if the loan is a term loan. All regressions include borrower × time × loan type and borrower × bank fixed effects. Columns 5-8 show the results using newly originated credits—both term loan and credit line—as the dependent variable. Standard errors are clustered at the bank level. Control variables include banks' NPL ratio, logarithm of total assets, ROA, Tier 1 capital ratio, and loan-asset ratio. Detailed variable definitions can be found in Appendix C. *p < .1; **p < .05; ***p < .01.

BANK FOR AMERICA: CREDIT LINE DURING COVID-19

Table 6: implications for bank lending

Understanding the mechanisms: Credit line repayments during 2020Q2-Q3 (post-policy interventions)

A. Repayment statistics in 2020Q2 and 2020Q3

A1. 2020Q2 repayments scaled by commitment

Rating	Mean	Median	SD	Min	Max
AAA-A	.24	.159	.266	.000	.990
BBB	.26	.199	.245	.000	1.000
Non-IG	.26	.149	.276	.000	1.000
NR	.2	.107	.243	.000	1.000

A2. 2020Q2 repayments scaled by remaining drawdown balance

Rating	Mean	Median	SD	Min	Max
AAA-A	.69	.988	.373	.000	1.000
BBB	.64	.736	.365	.000	1.000
Non-IG	.5	.409	.393	.000	1.000
NR	.39	.256	.369	.000	1.000

A3. 2020Q3 repayments scaled by commitment

Rating	Mean	Median	SD	Min	Max
AAA-A	.08	.000	.166	.000	.802
BBB	.12	.024	.184	.000	1.000
Non-IG	.15	.059	.208	.000	1.000
NR	.14	.068	.197	.000	1.000

A4. 2020Q3 repayments scaled by remaining drawdown balance

Rating	Mean	Median	SD	Min	Max
AAA-A	.57	.670	.395	.000	1.000
BBB	.52	.497	.382	.000	1.000
Non-IG	.43	.303	.380	.000	1.000
NR	.4	.286	.378	.000	1.000

B. Which banks recover from losses in market value?

	(1)	(2)	(3)	(4)	(5)	(6)
Fees earned	0.118*** (.010)	0.216*** (.003)	0.186*** (.000)	0.0772 (.299)	0.205*** (.009)	0.109* (.082)
Repayments	0.673*** (.046)	0.442 (.294)	-1.886*** (.004)	-0.195 (.493)	0.609 (.265)	-1.498** (.019)
MV loss COVID	0.499*** (.000)	0.561*** (.000)	0.294*** (.000)	0.565*** (.000)	0.569*** (.000)	0.388*** (.001)
Drawdowns 2020Q1	3.004*** (.006)	3.190*** (.006)	3.017*** (.004)	4.264*** (.000)	5.836* (.089)	5.038*** (.021)
Fees earned × Repayments			-0.382* (.061)	0.135 (.567)	-0.369 (.102)	-0.193 (.385)
Repayments × MV loss COVID			-0.562*** (.000)			1.889** (.016)
Repayments × Capital buffer				-31.25*** (.003)		-18.34* (.061)
Repayments × Drawdowns 2020Q1					-20.57 (.415)	-10.37 (.555)
Constant	-0.232** (.016)	-0.161 (.137)	0.144 (.106)	-0.120 (.215)	-0.183 (.133)	0.0564 (.569)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.901	.914	.951	.949	.917	.961
Number obs.	32	32	32	32	32	32

Panel A of Table 8 shows descriptive statistics for the repayment behavior of borrowers by rating category. Subpanels A1 and A2 display the behavior in 2020Q2 in relation to total committed credit or remaining drawdown balance, and subpanels A3 and A4 show the analog for 2020Q3. Panel B reports the results of OLS regressions of US banks' stock returns between March 23, 2020, and June 30, 2020, on bank-level variables capturing the opportunity-cost adjusted *Fees earned* on outstanding credit lines as well as credit-line *Repayments* during the 2020Q2 period. Columns 3, 4, and 5 interact *Repayments* with indicators of previous distress: market

Table 7: repayments and recovery

5.7 Table 8 and Table 9: other crises and credit-line pricing

Read:

- Table 8 shows that the credit-line channel is strongest in COVID. High commitments do not generate the same negative effect in pre-COVID crises.
- Table 9 studies whether drawdown-option risk is priced in AISD and AISU fees. Some systematic-risk variables are priced, but liquidity risk is not strongly priced.

Economic message: Banks earn fees for credit-line provision, but fees do not appear to fully internalize aggregate drawdown risk.

Table 8
Credit line commitments, liquidity risk, and bank stock returns during crises (including pre-COVID-19 crises)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Commitment above median	0.0166 (.144)	0.0144** (.045)		0.0143** (.010)	0.0143*** (.005)		0.00204 (.773)	0.00229 (.742)	
Crisis	-0.466*** (.000)	-0.253*** (.000)	-0.253*** (.000)	-0.0791*** (.000)	-0.0039 (.136)	-0.00343 (.725)	0.0683*** (.000)	0.0526*** (.000)	0.0536*** (.000)
Commitment above median $\times$ Crisis	-0.0811*** (.017)	-0.0786*** (.000)	-0.0784*** (.001)	-0.0002*** (.000)	-0.0299*** (.003)	-0.0308*** (.005)	-0.0117*** (.005)	-0.0118*** (.001)	-0.0333*** (.003)
MKTBF		-0.277* (.055)	-0.276 (.107)		0.367*** (.000)	0.366*** (.000)		0.201*** (.000)	0.205*** (.000)
SMB FF3		2.217*** (.000)	2.219*** (.000)		0.448*** (.000)	0.448*** (.000)		0.529*** (.000)	0.538*** (.000)
HML		0.549*** (.000)	0.351*** (.000)		1.099*** (.000)	1.112*** (.000)		0.540*** (.000)	0.547*** (.000)
Constant	0.117*** (.000)	0.0814*** (.000)	0.0919*** (.000)	-0.00325 (.489)	-0.0270*** (.000)	-0.0178*** (.000)	0.0272*** (.000)	0.0223 (.717)	0.0299 (.261)
Sample	COVID	COVID	COVID	GFC	GFC	GFC	Dot-com	Dot-com	Dot-com
Bank FE	No	No	Yes	No	No	Yes	No	No	Yes
R-squared	.485	.866	.806	.047	.221	.221	.024	.103	.103
Number obs.	1,364	1,364	1,364	8,109	8,109	8,109	3,914	3,914	3,914

This table reports the results of OLS regressions of quarterly U.S. banks' excess stock returns for three samples on a dummy variable indicating banks with above median credit line commitments (assigned one quarter before the respective crisis), a dummy variable indicating a crisis quarter and control variables. Separate time-series samples are 2010Q4 to 2013Q1 (COVID), 2008Q1 to 2011Q4 (GFC), and 2000Q1 to 2002Q4 (dot-com). Crisis quarters are 2001Q1 to 2001Q4 (dot-com), 2007Q3 to 2009Q2 (GFC), and 2020Q1 (COVID). Columns sequentially add control variables and bank fixed effects for each sample. The sample of banks is matched on total assets, capitalization, NPL, no-loans, and loans-to-assets ratios. Appendix C defines all variables. *p < .1; **p < .05; ***p < .01 (based on robust standard errors).

The Review of Financial Studies / V 27 / n 2 2024

Table 8: credit-line commitments across crises

Pricing of drawdown options in credit line fees

A. AISD

	(1)	(2)	(3)	(4)	(5)
	AISD				
<i>Bank equity beta</i>		0.0582 (.147)			
<i>LRMES</i>			1.293** (.039)		
<i>SRISK / assets</i>				1.772 (.293)	
<i>Liquidity risk</i>					-0.330 (.185)
<i>LIBOR</i>	-0.288*** (.000)	-0.278*** (.001)	-0.243*** (.006)	-0.272*** (.002)	-0.311*** (.000)
<i>Bank characteristics</i>					
<i>NPL / loans</i>	1.864 (.342)	2.298 (.261)	2.120 (.281)	2.474 (.242)	1.832 (.339)
<i>Capital</i>	-5.395** (.013)	-4.925** (.018)	-4.967** (.013)	-4.981** (.020)	-5.412*** (.009)
<i>Noninterest income</i>	-0.0560 (.796)	-0.0490 (.820)	-0.171 (.458)	-0.0628 (.773)	-0.163 (.476)
<i>Bank size</i>	-0.107*** (.001)	-0.111*** (.001)	-0.110*** (.001)	-0.119*** (.001)	-0.126*** (.000)
<i>Bank profitability</i>	-10.20** (.042)	-8.032 (.113)	0.132 (.984)	-2.877 (.741)	-10.47** (.036)
<i>Firm characteristics</i>					
<i>Equity volatility</i>	0.360** (.019)	0.366** (.015)	0.367** (.015)	0.364** (.016)	0.360** (.020)
<i>Firm equity beta</i>	0.175*** (.001)	0.176*** (.001)	0.173*** (.001)	0.174*** (.001)	0.176*** (.000)
<i>Firm size</i>	-0.170*** (.000)	-0.169*** (.000)	-0.167*** (.000)	-0.169*** (.000)	-0.170*** (.000)
<i>Firm profitability</i>	-0.200 (.327)	-0.201 (.328)	-0.193 (.345)	-0.203 (.318)	-0.211 (.290)
<i>Tangibility</i>	-0.475*** (.000)	-0.478*** (.000)	-0.475*** (.000)	-0.474*** (.000)	-0.476*** (.000)
<i>Tobin's q</i>	-0.0279** (.040)	-0.0281** (.038)	-0.0274** (.042)	-0.0275** (.042)	-0.0265** (.049)
<i>Leverage</i>	1.756*** (.000)	1.753*** (.000)	1.764*** (.000)	1.755*** (.000)	1.757*** (.000)
<i>Year fixed effect</i>	Yes	Yes	Yes	Yes	Yes
<i>Industry fixed effect</i>	Yes	Yes	Yes	Yes	Yes
<i>R-squared</i>	.463	.463	.464	.463	.463
<i>Number obs.</i>	2657	2657	2657	2657	2657

(Continued)

(Continued)

B. AISU

	(1)	(2)	(3)	(4)	(5)
	AISU				
<i>Bank equity beta</i>		0.0161** (.021)			
<i>LRMES</i>			0.187* (.085)		
<i>SRISK / assets</i>				0.255 (.382)	
<i>Liquidity risk</i>					-0.0253 (.581)
<i>LIBOR</i>	-0.0516*** (.000)	-0.0488*** (.000)	-0.0451*** (.002)	-0.0492*** (.001)	-0.0534*** (.000)
<i>Bank characteristics</i>					
<i>NPL / loans</i>	0.565* (.072)	0.684** (.032)	0.602* (.054)	0.652* (.053)	0.562* (.072)
<i>Capital</i>	-0.576 (.153)	-0.446 (.246)	-0.514 (.173)	-0.516 (.198)	-0.577 (.146)
<i>Noninterest income</i>	0.0103 (.795)	0.0122 (.752)	-0.00638 (.880)	0.00930 (.815)	0.00208 (.962)
<i>Bank size</i>	-0.0117** (.034)	-0.0127** (.019)	-0.0122** (.022)	-0.0135** (.025)	-0.0132** (.040)
<i>Bank profitability</i>	-1.486* (.077)	-0.887 (.300)	0.0100 (.993)	-0.430 (.765)	-1.507* (.074)
<i>Firm characteristics</i>					
<i>Equity volatility</i>	0.0700*** (.000)	0.0716*** (.000)	0.0709*** (.000)	0.0706*** (.000)	0.0700*** (.000)
<i>Firm equity beta</i>	0.0482*** (.000)	0.0485*** (.000)	0.0480*** (.000)	0.0480*** (.000)	0.0482*** (.000)
<i>Firm size</i>	-0.0326*** (.000)	-0.0324*** (.000)	-0.0323*** (.000)	-0.0325*** (.000)	-0.0327*** (.000)
<i>Firm profitability</i>	0.0113 (.716)	0.0110 (.725)	0.0123 (.693)	0.0108 (.726)	0.0105 (.735)
<i>Tangibility</i>	-0.0904*** (.000)	-0.0913*** (.000)	-0.0905*** (.000)	-0.0903*** (.000)	-0.0905*** (.000)
<i>Tobin's q</i>	-0.0103*** (.000)	-0.0104*** (.000)	-0.0103*** (.000)	-0.0103*** (.000)	-0.0102*** (.000)
<i>Leverage</i>	0.320*** (.000)	0.319*** (.000)	0.321*** (.000)	0.320*** (.000)	0.320*** (.000)
<i>Year fixed effect</i>	Yes	Yes	Yes	Yes	Yes
<i>Industry fixed effect</i>	Yes	Yes	Yes	Yes	Yes
<i>R-squared</i>	.472	.473	.473	.472	.472
<i>Number obs.</i>	2,657	2,657	2,657	2,657	2,657

This table reports the results of OLS regressions of the All-In-Spread-Drawn (AISD) in panel A and the All-In-Spread-Undrawn (AISU) in panel B on banks' aggregate risk exposures including *Bank equity beta* (as a measure of systematic risk), *LRMES* (as a measure of downside risk; LRMES is the Long-Run Marginal Expected Shortfall, approximated in Acharya, Engle, and Richardson (2012) as $1 - e^{-(18 \times \text{MES})}$), where MES is the 1-day loss expected in a bank's return if market returns are less than -2%, *SRISK/Assets* (as a measure of equity shortfall in times of a severe crisis) and *Liquidity risk* (as a measure of aggregate drawdown risk and defined as Unused Commitments plus Wholesale Funding minus Liquidity (% Assets)). We include them individually in regressions (2) to (5) and (7) to (10). All regressions include bank characteristics: *NPL/Loans* (Nonperforming loans (% loans)), *Capital* (Equity/Assets), *Noninterest income* (Noninterest income (% Operating revenues)),

Table 9, panel A: AISD

Table 9, panel B: AISU

5.8 Figure 5, Figure 6, and Table 10: issuance, drawdown slopes, and stress tests

Read:

- Figure 5 shows that credit-line issuance temporarily falls after COVID but later recovers; fee behavior is mixed.
- Figure 6 shows that COVID drawdowns are more sensitive to market downturns than GFC drawdowns.
- Table 10 predicts a 22.91% drawdown rate under a 40% market correction using COVID sensitivity, versus 10.82% using GFC sensitivity.
- The incremental SRISK from credit-line drawdowns is large for the largest banking organizations.

Economic message: Stress testing should treat credit lines as contingent exposures whose conversion risk is state dependent and much larger in COVID-style market freezes.

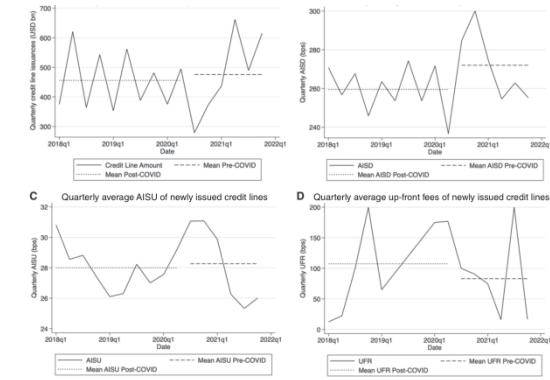


Figure 5
Credit line issuances (volume and spread/fees)
This figure shows quarterly issuance volume in USD billion (panel A), all-in-spread-drawn or AISD (panel B), all-in-spread-undrawn or AISU (panel C), and up-front fees or UFR (panel D), with spreads and fees in basis points (bps), of credit line issuances by U.S. firms over the 2018 to 2021 period. All variables are defined in Appendix C.

Figure 5: credit-line issuance and fees

Credit-line drawdowns and conditional SRISK

A. Estimating the drawdown rates in a stress scenario

Predicted Drawdowns	Quarterly	Quarterly	Q1 2020	2007–2009	Slope	Drawdown rate (S&P return = 40%)
					–0.57	22.91%
					–0.27	10.82%

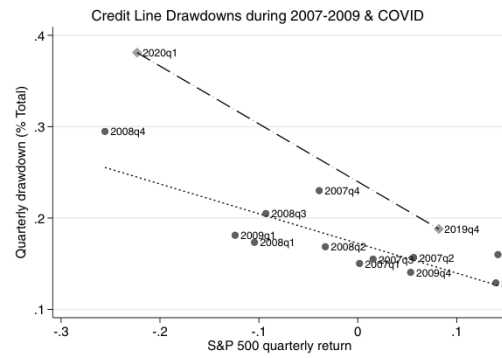
B. Incremental SRISK^{CL}

Name	Unused C&I commitments (USD mn)	Incremental SRISK ^{CL} with drawdown rate: 25.79%	Incremental SRISK ^{CL} with drawdown rate: 39.97%	Debt (USD mn)
JPMORGAN CHASE & CO.	273,278	5,638	8,738	2,496,125
BANK OF AMERICA CORPORATION	310,824	6,413	9,939	2,158,067
CITIGROUP INC.	200,912	4,145	6,424	1,817,838
WELLS FARGO & COMPANY	198,316	4,092	6,341	1,748,234
GOLDMAN SACHS GROUP INC., THE	111,247	2,295	3,557	913,472
MORGAN STANLEY	78,411	1,618	2,507	818,732
U.S. BANCORP	96,020	1,981	3,070	433,158
TRUIST FINANCIAL CORPORATION	86,995	1,795	2,782	204,178
PNC FINANCIAL SERVICES GROUP INC., THE	84,238	1,738	2,694	358,342
CAPITAL ONE FINANCIAL CORPORATION	18,618	384	595	320,520
Top-10 BHC	1,458,858	30,099	46,648	11,268,666
Vlab BHC	1,777,617	36,676	56,841	14,524,200
All BHC	1,837,220	37,906	58,747	

This table reports the predicted drawdown rates (*Drawdown rate*) from credit lines in a stress scenario of 40% correction to the global stock market (panel A) and the *Slope* of the drawdown function (compare Figure 6). In panel B, we report the *Unused commitments* (C&I loans), and the incremental required capital to fund the predicted drawdowns (*Incremental SRISK^{CL}*) using both (stressed) historical drawdown rates: $\text{Incremental SRISK}^{\text{CL}} = \text{Drawdown rate} \times 8\% \times \text{Unused commitments}$ (C&I loans). *Debt* is total liabilities (from NYU Stern School of Business VLAB site, vlab.stern.nyu.edu/srisk). Panel C reports the calculation of *Incremental SRISK^{LRMES-C}* due to the sensitivity of bank stock returns to *Liquidity risk* using the minimum ($\gamma_{\min}$) and maximum ($\gamma_{\max}$) sensitivity from different model specifications shown in prior tables. $\text{Incremental LRMEs-C}_{\min}$ (%) is calculated as $\text{Liquidity Risk} \times \gamma_{\min}$. $\text{Incremental SRISK}^{\text{LRMES-C}_{\min}}$ is calculated as $(1\% - 8\%) \times \text{Liquidity risk} \times \gamma_{\min} \times \text{MV}$, where MV is market value of bank equity. Other variants are calculated accordingly. In panel D, we show the Conditional SRISK (*SRISK-C*), which is the

Table 10, panels A–B

A Quarterly drawdowns versus quarterly S&P 500 returns



B Quarterly drawdowns versus lowest cumulative S&P 500 return in each quarter

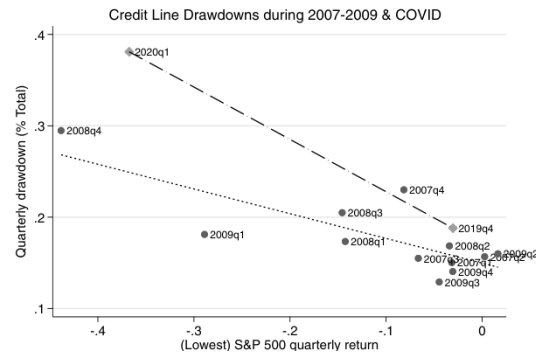


Figure 6: drawdowns and market returns

Table 10

(Continued)

C. Incremental SRISK^{LRMES-C}

	MV (USD mn)	LRMES	Liquidity risk	$\gamma_{\min}$	$\gamma_{\max}$	Incremental LRMEs-C _{min}	Incremental LRMEs-C _{max}	Incremental SRISK ^{LRMES-C} (USD mn)	Incremental SRISK ^{LRMES-C} (USD mn)
JPMORGAN CHASE & CO.	417,226	43.4%	20.3%	-0.32	-0.56	6.9%	11.3%	28,411	49,276
BANK OF AMERICA CORPORATION	316,808	45.9%	25.7%	-0.32	-0.56	8.2%	14.3%	26,052	45,183
CITIGROUP INC.	174,415	47.3%	37.1%	-0.32	-0.56	11.9%	20.6%	20,698	38,883
WELLS FARGO & COMPANY	227,540	44.9%	24.2%	-0.32	-0.56	7.7%	13.4%	17,612	30,346
GOLDMAN SACHS GROUP INC., THE	81,415	54.2%	28.7%	-0.32	-0.56	8.2%	15.9%	7,471	12,908
MORGAN STANLEY	82,743	51.1%	14.3%	-0.32	-0.56	4.6%	7.9%	3,781	6,357
U.S. BANCORP	92,405	46.7%	46.7%	-0.32	-0.56	14.6%	25.7%	13,780	23,613
TRUIST FINANCIAL CORPORATION	75,344	42.5%	41.1%	-0.32	-0.56	13.2%	22.8%	9,943	17,245
PNC FINANCIAL SERVICES GROUP INC., THE	69,045	40.1%	39.9%	-0.32	-0.56	12.8%	22.1%	8,628	15,485
CAPITAL ONE FINANCIAL CORPORATION	47,927	49.2%	18.6%	-0.32	-0.56	5.9%	10.3%	2,849	4,942
Top-10 BHC	1,606,166					9.5%	16.4%	139,467	241,888
Vlab BHC	2,250,522							166,438	262,134
All BHC	2,408,434							177,412	307,699

D. SRISK^C (USD mn)

Name	w/req SRISK	w/req SRISK	SRISK-C _{min}	SRISK-C _{max}
JPMORGAN CHASE & CO.	0	-27,848	14,095	58,014
BANK OF AMERICA CORPORATION	14,898	-14,898	25,465	55,122
WELLS FARGO & COMPANY	24,425	-24,425	21,704	36,887
CITIGROUP INC.	60,887	-60,887	24,835	42,788
AMERICAN EXPRESS COMPANY	0	-35,344	5,688	9,864
U.S. BANCORP	0	-15,352	15,711	26,883
MORGAN STANLEY	28,302	28,302	5,368	9,964
GOLDMAN SACHS GROUP INC., THE	38,774	38,774	9,766	16,543
TRUIST FINANCIAL CORPORATION	0	-23,608	11,738	20,026
PNC FINANCIAL SERVICES GROUP INC., THE	0	-15,995	10,666	18,179
Top-10 (w/banks)	167,287	51,228	172,020	292,863
Total (Vlab banks)	195,033	48,994	205,113	348,975
Total (all sample banks)			815,312	1,496,496

Downloaded from https://academic.oup.com/rfs/advance-article/doi/10.1093/rfs/fhza010 by Australian National University user on 04 May 2020

Table 10, panels C–D

6 Short summary

- Banks expanded credit-line commitments substantially before COVID.
- COVID created a sudden aggregate demand for liquidity, causing firms to draw credit lines aggressively.
- Bank stock prices crashed more than nonbank financials and nonfinancial firms.

- Banks with greater balance-sheet liquidity risk had larger stock-price declines.
- The effect is robust to controls for bank fundamentals, SRISK, and loan exposure to COVID-affected industries.
- The timing is episodic: liquidity risk matters most before decisive policy interventions.
- Gross drawdowns, not net drawdowns, explain bank stock losses, implying a capital-encumbrance channel.
- Deposit inflows made the funding-liquidity channel less central during COVID.
- Higher capital buffers mitigate the negative effect of drawdowns.
- Drawdown-exposed banks reduce new loan originations, showing that credit-line stress affects intermediation.
- Repayments release capital and help explain bank stock recoveries in 2020Q2–Q3.
- Credit-line fees partially price risk but do not appear to fully price aggregate drawdown risk.
- The paper proposes adding conditional credit-line drawdown risk to bank stress tests.

7 Conclusion

7.1 Core conclusion

Acharya, Engle, Jager, and Steffen show that bank credit lines are not just liquidity insurance for borrowers; they are also contingent liabilities for banks. During COVID, the aggregate drawdown state arrived suddenly. Banks with greater exposure to unused credit lines lost more market value because those commitments became capital-absorbing loans.

7.2 Economic intuition

A credit line is like a put option written by the bank. The borrower exercises the option when market funding freezes and liquidity is most valuable. If many borrowers exercise simultaneously, the bank's balance sheet expands precisely when capital is scarce. Deposit inflows may cover the funding need, but they do not eliminate the capital cost of drawn loans. Repayments later reverse this balance-sheet pressure.

7.3 Takeaway

The paper reframes credit lines as a source of systemic bank fragility. The correct regulatory lesson is not simply to monitor deposits or wholesale funding, but to stress-test off-balance-sheet commitments under aggregate drawdown scenarios. In a crisis, unused commitments can become real loans, and the stock market prices that conversion risk.